Machine Learning A (2026)
Home Assignment 7

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1 Conceptual Questions

1. **False**: but in expectation it is true. That is for the output constructed using k-means++, the corresponding cost function satisfies[1]

$$\mathbb{E}[\phi] \leq 8(\log k + 2)\phi^{\star}, \tag{1}$$

where $\phi^{\star}$ is the lowest possible k-means cluster cost. (1) does not guarantee that any particular initialization resulting in ϕ_0 will fulfill $\phi_0 \leq 8(\log k + 2)\phi^{\star}$ only in expectation.

2. k-means clustering and PCA are both *unsupervised learning* methods and thus somehow try to extract patterns for understanding and or visualisation of the data. Typically in a data-exploratory state. One way they differ is in their concrete goal. The goal of PCA is to linearly reduce the dimensions needed to represent the information / variation in the data (or as much as possible). The goal of k-means clustering is to identify datapoints $x \in \mathbb{R}^m$ that are share some similarities.

2 k-means vs. k-means++

2.1 Part 1: k-means++ theory

Generally for k-means++ the probability that $x \in \mathcal{X}$ is chosen as c_i $(i < k)$ when initial clustering centers $S = \{c_{1,2,\ldots,i-1}\}$ are already chosen is

$$\mathbb{P}(c_i = x \mid \{c_{1,2,\ldots,i-1}\}) = \frac{D(x, \{c_{1,2,\ldots,i-1}\})^2}{\sum_{u \in \mathcal{X}} D(u, \{c_{1,2,\ldots,i-1}\})^2}, \tag{2}$$

where

$$D(x, \{c_{1,2,\ldots,i-1}\}) = \min_{z \in \{c_{1,2,\ldots,i-1}\}} ||x - z||_2$$

Therefore in the *small-dataset* with $c_1 = A = (-2, 2)$ and $c_2 = D = (7, -5)$ already we have (where i only calculate visually ambiguous distances)

$$\begin{aligned} D(B, \{A, D\}) &= D((0,4), \{(-2,2), (7,-5)\}) \\ &= ||(0,4) - (-2,2)|| \\ &= \sqrt{8} \end{aligned}$$

$$\begin{aligned} D(C, \{A, D\}) &= D((6,3), \{(-2,2), (7,-5)\}) \\ &= \min\{||(6,3) - (-2,2)||_2, ||(6,3) - (7,-5)||_2\} \\ &= \min\left\{\sqrt{65}, \sqrt{65}\right\} \\ &= \sqrt{65} \end{aligned}$$

[1] Sadegh Talebi, Clustering: k-means and Beyond, slide 33

$$
\begin{aligned}
D\left(E, \{A, D\}\right) &= D\left((4,-6), \{(-2,2),(7,-5)\}\right) \\
&= ||(4,-6) - (7,-5)||_2 \\
&= \sqrt{10}
\end{aligned}
$$

$$
\begin{aligned}
D\left(F, \{A, D\}\right) &= D\left((0,-4), \{(-2,2),(7,-5)\}\right) \\
&= \min\left\{||(0,-4) - (-2,2)||_2, ||(0,-4) - (7,-5)||_2\right\} \\
&= \min\left\{\sqrt{40}, \sqrt{50}\right\} \\
&= \sqrt{40}.
\end{aligned}
$$

From this we simply apply (2) yielding the following probability distribution

$$
\begin{aligned}
\mathbb{P}\left(c_3 = B\right) &= \frac{8}{123} \\
\mathbb{P}\left(c_3 = C\right) &= \frac{65}{123} \\
\mathbb{P}\left(c_3 = E\right) &= \frac{10}{123} \\
\mathbb{P}\left(c_3 = F\right) &= \frac{40}{123}.
\end{aligned}
$$

Note that strictly speaking (2) also has $D\left(A, \{A, D\}\right) + D\left(D, \{A, D\}\right) = 0$ in the denominator.

2.2 Part 2: Comparing seeding methods for k-means clustering

Using the assignments initialization of $(0, 4), (1, 5)$ and $(-1, 5.5)$ and sklearn's implementation of k-means clustering i obtain the following final cluster centers $(3.3, -5.24), (6, 3)$ and $(-0.85, 3.73)$ also displayed in figure 1.

To examine the effects of the initial cluster i first sample 40 random initializations. Then i record the lowest empirical k-means cost obtained and count how many obtains this score (to computer precision).

As a reminder the k-means cost is computed as

$$
\phi(\mathcal{X}, \{c_1, c_2, c_3\}) \equiv \sum_{x \in \mathcal{X}} \min_{c \in \{c_1, c_2, c_3\}} ||x - c||_2^2.
$$

I find the optimal cluster centers to be $(-0.01, 3.64), (5.24, -6.03)$ and $(0.2, -3.98)$ (see figure 2) with a k-means cost of 93.67 which was reached 15 out of 40 times.

Using in stead k-means++ for initialization and running this 40 times i obtain the same optimal clustering center new 31 out of 40 times.

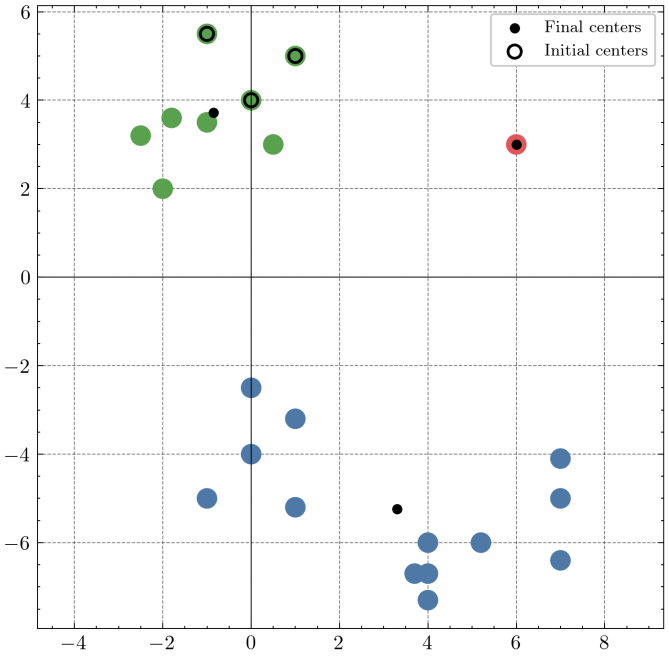


Figure 1: Cluster centers found by k-means clustering ($k = 3$)

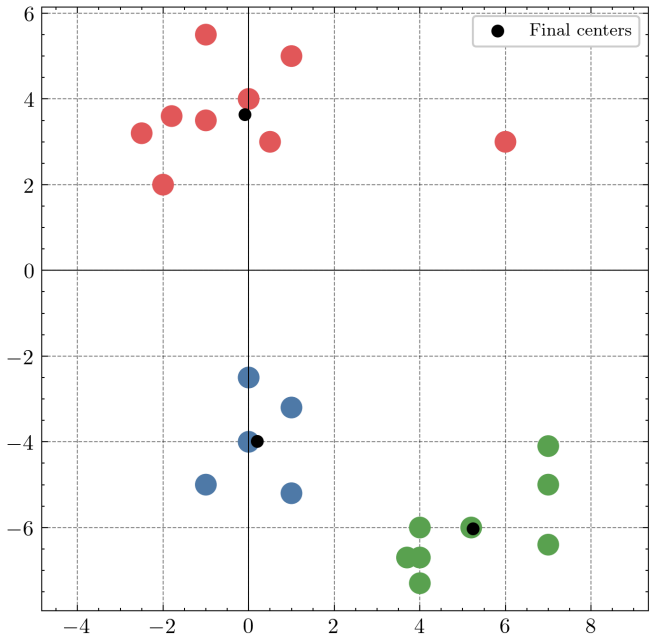


Figure 2: Lowest k-means cost clustering centers found ($k = 3$)

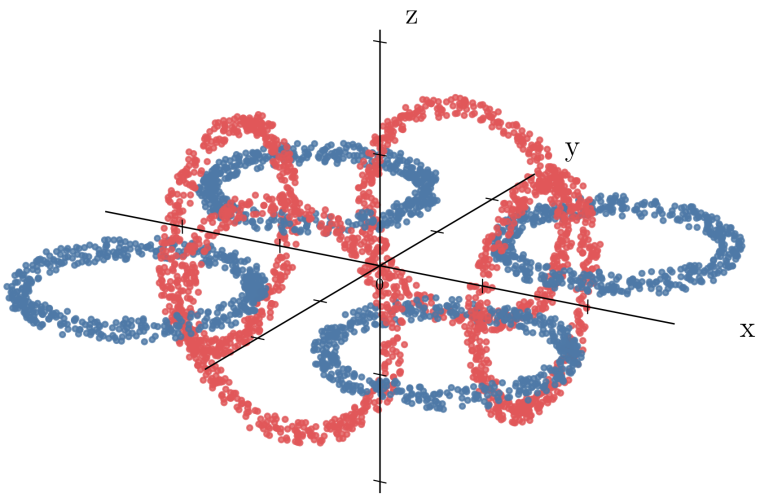


Figure 3: Visualization of “vis-dataset”

3 Visualization

3.1 Plotting

The dataset is plottet in figure 3 as described in the assignemnt i color the dots based on what half of the dataset they reside.

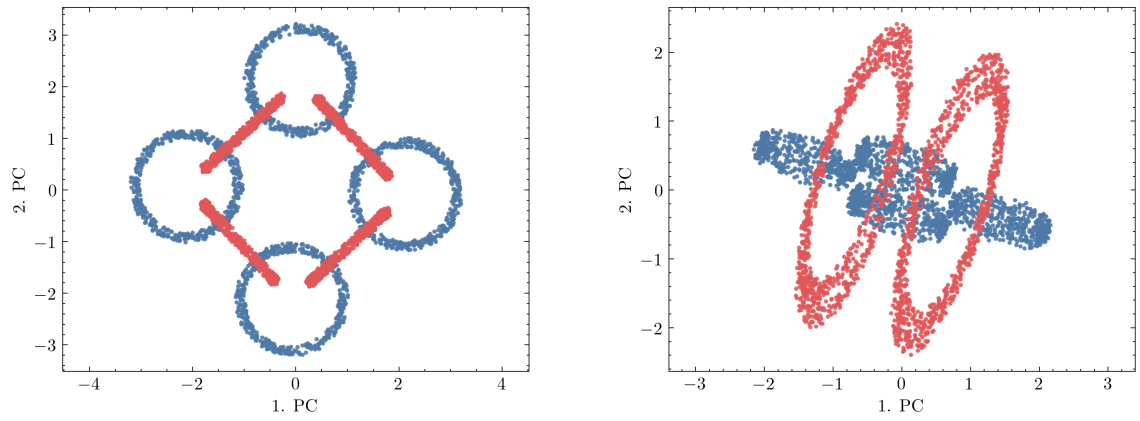
3.2 PCA

I use the sklearn implementation of PCA and their scaler to standardise the data (i.e. mean zero and unit variance of each dimension). Figure 4 displays the recreated data from the resulting first two principal components. Panel (a) shows the results for the raw data and panel (b) for standardized.

This highlights the potential effects of prepossessing on PCA. PCA is sensitive to variance of each dimension in the data and because its based on the squared error between recreated / embedded datapoints and original points in $\mathcal{X}$ the results are sentitive to scaling.

3.3 t-SNE

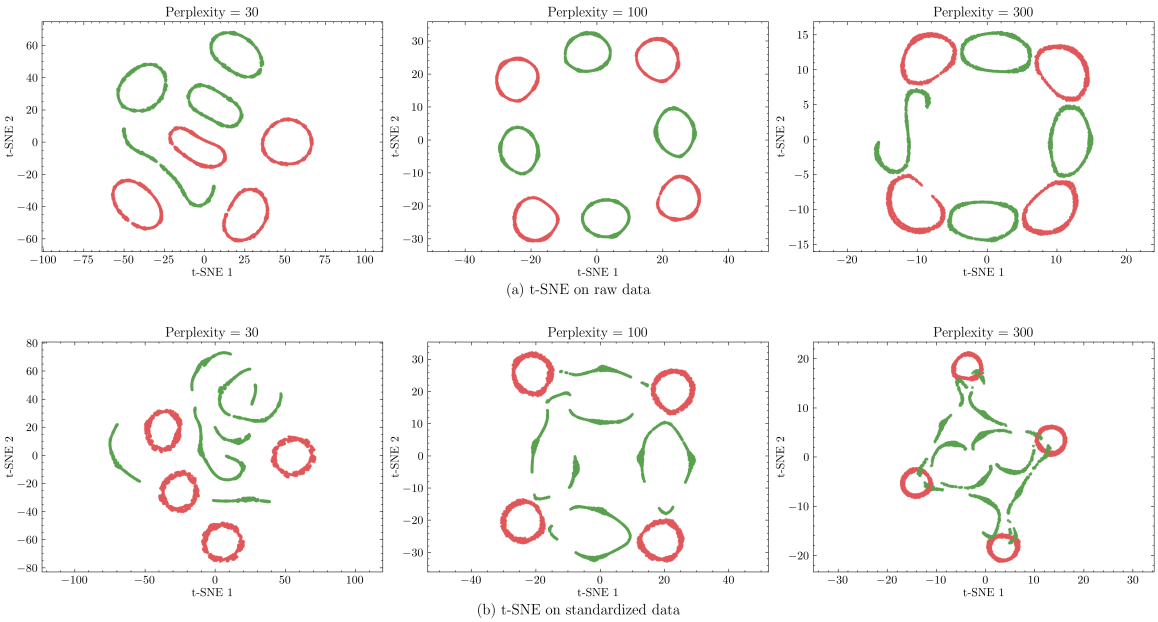
Again, I use the implementation in sklearn to fit t-SNE. I use a random initialization with a fixed seed and gradient descent to find a local minimum of the non-convex KL-divergence loss. Figure 5 presents the results for both raw and standardized data, using perplexities of 30, 100, and 300.



(a) PCA on raw data

(b) PCA on standardized data

Figure 4: Data recreation from first two PC



(a) t-SNE on raw data

(b) t-SNE on standardized data

Figure 5: t-SNE results for different choices of perplexity

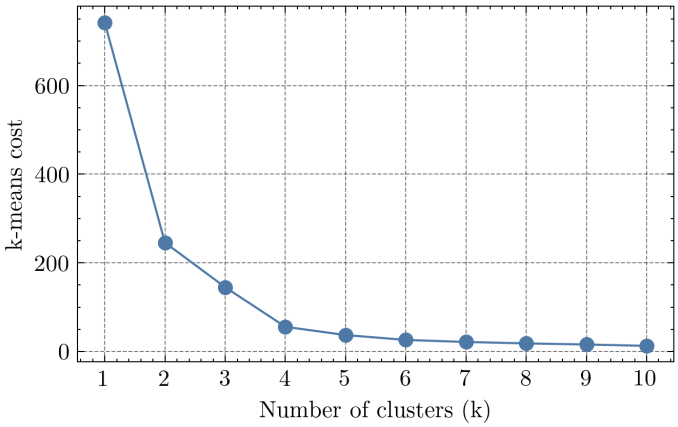


Figure 6: k-means loss over k
Note: $k > 10$ is not plottet as they are not relevant for the elbow method in this case.

For the raw data, t-SNE reveals several distinct circular structures. These correspond to different loop-shaped regions of the two intertwined point clouds in the original three-dimensional data, rather than necessarily representing separate clusters. At perplexity 30, some loops are distorted or broken, while perplexity 300 compresses the structures and brings them closer together. I would therefore choose perplexity 100, which produces the clearest separation while preserving the approximately circular shapes.

For the standardized data, the structures become less clearly separated, particularly at perplexity 300, where several loops overlap. This illustrates how standardization changes the neighborhood relationships that t-SNE attempts to preserve.

Compared with PCA, t-SNE better reveals the local nonlinear structure of the data. However, the relative positions and distances between the resulting loops cannot be interpreted as accurately reflecting their geometry in the original three-dimensional space.

4 Determining the Number of Clusters

The relevant non-trivial choices of k are $k \in \{2, 3, \ldots, 21\}$ where $k = 1$ assigns all points to one cluster (thus c_1 will be the centroid) and $k = 22$ assigns one c_i to each point x_i where $c_i = x_i$.

To determine the appropriate number of clusters i use the heuristic method of finding the *elbow point*. This is the k whereafter the k-means cost stops decreasing "rapidly". Therefore we strike a kind of balance between parameters and loss.

Based on figure 6 $k = 3$ or $k = 4$ seem like reasonable choices.